

INPUTS FOR EQUIVARIANT INFINITE LOOP SPACE MACHINE

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ABSTRACT. Symmetric monoidal categories are central studies in modern algebraic topology, for instance K -theory relies on constructing spectra with suitable homotopy groups out of symmetric monoidal categories using the infinite loop space machine. It's known that the two inputs - pseudoalgebra over an operad $\mathcal{P}$, or suitable categories described by based finite sets $\mathcal{F}$ - are equivalent description for symmetric monoidal categories.

Within equivariant K -theory, similar machineries have been established, where the input for constructing G -spectra are described by pseudoalgebra over an operad $\mathcal{P}_G$ as G -categories, or categories described by based finite G -sets $\mathcal{F}_G$. The equivalence of the two within equivariant case is a natural question to ask, and associates to the question of “what is symmetric monoidal G -categories”.

We'll provide an exposition on the two inputs by collecting the axioms scattered (or implicitly assumed) in various literatures, while describe some equivariance properties of the two inputs.

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1. CATEGORICAL PRELIMINARIES

1.1. 2-Categories.

Definition 1.1. A category $\mathcal{C}$ is a *2-category*, if for each object $X, Y \in \mathcal{C}$ (called *0-cells*), the homset $\mathcal{C}(X, Y)$ is itself a category (called the *hom-category*). Also, for every 0-cells $X, Y, Z \in \mathcal{C}$, the *composition* $\circ : \mathcal{C}(Y, Z) \times \mathcal{C}(X, Y) \rightarrow \mathcal{C}(X, Z)$ is a functor.

Every morphism $f \in \mathcal{C}(X, Y)$ is called a *1-cell*, while given any 1-cells $f, g \in \mathcal{C}(X, Y)$, a morphism $\alpha : f \Rightarrow g$ in the hom-category $\mathcal{C}(X, Y)$ is called a *2-cell*.

Here, $\circ$ must be strictly unital and associative.

Remark 1.2. There is a more general notion of a *bicategory*, which relaxes the composition functors to be associative and unital up to coherent natural isomorphisms, and in some literature this is called 2-categories. Here we'll take the strict associativity and unitality as axioms for 2-categories. For more general bicategory, cf. [?] Section 2.1.

Example 1.3. Let **Cat** denote the *category of small categories*, where the objects are small categories, the hom-categories **Cat**($\mathcal{A}, \mathcal{B}$) are functors between small categories $\mathcal{A}$ and $\mathcal{B}$, and the morphisms in **Cat**($\mathcal{A}, \mathcal{B}$) are natural transformations between functors. This specifies **Cat** to be a 2-category.

1.2. Pseudofunctor, Pseudotransform, and Modification.

Definition 1.4 (Pseudofunctor). Given two 2-categories $\mathcal{A}, \mathcal{B}$, a *pseudofunctor* $F : \mathcal{A} \rightarrow \mathcal{B}$ equips with an 0-cell map, functors between hom-categories $F_{X,Y} : \mathcal{A}(X, Y) \rightarrow \mathcal{B}(F(X), F(Y))$, an invertible 2-cell $\phi_{X,Y,Z}$ between compositions

$$\begin{array}{ccc}
 \mathcal{A}(Y, Z) \times \mathcal{A}(X, Y) & \xrightarrow{F_{Y,Z} \times F_{X,Y}} & \mathcal{B}(F(Y), F(Z)) \times \mathcal{B}(F(X), F(Y)) \\
 \downarrow \circ & \nearrow \phi_{X,Y,Z} & \downarrow \circ \\
 \mathcal{A}(X, Z) & \xrightarrow{F_{X,Z}} & \mathcal{B}(F(X), F(Z))
 \end{array}$$

and another invertible 2-cell ψ_X between identities

$$\begin{array}{ccc}
 & \text{id}_{F(X)} & \\
 F(X) & \begin{array}{c} \curvearrowright \\ \downarrow \psi_X \\ \curvearrowleft \end{array} & F(X) \\
 & F_{X,X}(\text{id}_X) &
 \end{array}$$

that satisfies:

(i) **(Associativity)** The following two pasting diagrams are equal:

$$\begin{array}{ccc}
 \mathcal{A}(Z, W) \times \mathcal{A}(Y, Z) \times \mathcal{A}(X, Y) & \xrightarrow{\Pi F_{--}} & \mathcal{B}(F(Z), F(W)) \times \mathcal{B}(F(Y), F(Z)) \times \mathcal{B}(F(X), F(Y)) \\
 \downarrow \circ \times \text{id} & \swarrow \phi_{Y, Z, W} \times \text{id} & \downarrow \circ \times \text{id} \\
 \mathcal{A}(Y, W) \times \mathcal{A}(X, Y) & \xleftarrow{F_{Y, W} \times F_{X, Y}} & \mathcal{B}(F(Y), F(W)) \times \mathcal{B}(F(X), F(Y)) \\
 \downarrow \circ & \swarrow \phi_{X, Y, W} & \downarrow \circ \\
 \mathcal{A}(X, W) & \xleftarrow{F_{X, W}} & \mathcal{B}(F(X), F(W))
 \end{array}$$

equals

$$\begin{array}{ccc}
 \mathcal{A}(Z, W) \times \mathcal{A}(Y, Z) \times \mathcal{A}(X, Y) & \xrightarrow{\Pi F_{--}} & \mathcal{B}(F(Z), F(W)) \times \mathcal{B}(F(Y), F(Z)) \times \mathcal{B}(F(X), F(Y)) \\
 \downarrow \text{id} \times \circ & \swarrow \text{id} \times \phi_{X, Y, Z} & \downarrow \text{id} \times \circ \\
 \mathcal{A}(Z, W) \times \mathcal{A}(X, Z) & \xleftarrow{F_{Z, W} \times F_{X, Z}} & \mathcal{B}(F(Z), F(W)) \times \mathcal{B}(F(X), F(Z)) \\
 \downarrow \circ & \swarrow \phi_{X, Z, W} & \downarrow \circ \\
 \mathcal{A}(X, W) & \xleftarrow{F_{X, W}} & \mathcal{B}(F(X), F(W))
 \end{array}$$

(ii) **(Unitality)** The following two unit diagrams commute:

$$\begin{array}{ccc}
 \mathcal{A}(X, Y) \times \mathcal{A}(X, X) & \xrightarrow{F_{X, Y} \times F_{X, X}} & \mathcal{B}(F(X), F(Y)) \times \mathcal{B}(F(X), F(X)) \\
 \uparrow \text{id} \times \text{id}_X & \swarrow \text{id} * \psi_X & \downarrow \circ \\
 \mathcal{A}(X, Y) \times * & & \\
 \downarrow F_{X, Y} \times \text{id}_{F(X)} & \swarrow & \downarrow \circ \\
 \mathcal{B}(F(X), F(Y)) \times \mathcal{B}(F(X), F(X)) & \xrightarrow{\circ} & \mathcal{B}(F(X), F(Y))
 \end{array}$$

$$\begin{array}{ccc}
 \mathcal{A}(Y, Y) \times \mathcal{A}(X, Y) & \xrightarrow{F_{Y, Y} \times F_{X, Y}} & \mathcal{B}(F(Y), F(Y)) \times \mathcal{B}(F(X), F(Y)) \\
 \uparrow \text{id}_Y \times \text{id} & \swarrow \psi_Y * \text{id} & \downarrow \circ \\
 * \times \mathcal{A}(X, Y) & & \\
 \downarrow \text{id}_{F(Y)} \times F_{X, Y} & \swarrow & \downarrow \circ \\
 \mathcal{B}(F(Y), F(Y)) \times \mathcal{B}(F(X), F(Y)) & \xrightarrow{\circ} & \mathcal{B}(F(X), F(Y))
 \end{array}$$

where $\text{id}_X : * \rightarrow \mathcal{A}(X, X)$ is the functor indicating identity of X .

If ψ_X is identity 2-cell for all 0-cell X , then the pseudofunctor F is called *normal*.

Definition 1.5 (Pseudotransform). Fix 2-categories $\mathcal{A}, \mathcal{B}$, let $(F, \phi, \psi), (G, \varphi, \vartheta) : \mathcal{A} \rightarrow \mathcal{B}$ be two pseudofunctors. A *pseudotransform* $\mu : (F, \phi, \psi) \rightarrow (G, \varphi, \vartheta)$

collects 1-cells $\mu_X : F(X) \rightarrow G(X)$ for each 0-cell $X \in \mathcal{A}$, and for each 1-cell $f : X \rightarrow Y$ in $\mathcal{A}$, associates a 2-cell μ_f such that

$$\begin{array}{ccc} F(X) & \xrightarrow{\mu_X} & G(X) \\ Ff \downarrow & \mu_f \swarrow \searrow & \downarrow Gf \\ F(Y) & \xrightarrow{\mu_Y} & G(Y) \end{array}$$

commutes. They also satisfy:

- (i) (**Associativity**) Let $f : X \rightarrow Y$, $g : Y \rightarrow Z$ be 1-cells in $\mathcal{A}$, the following two pasting diagrams are equal:

$$\begin{array}{ccc} F(X) & \xrightarrow{\mu_X} & G(X) \\ \downarrow Ff & \searrow \mu_f & \downarrow Gf \\ F(Y) & \xrightarrow{\mu_Y} & G(Y) \\ \downarrow F(g \circ f) & \swarrow \phi_{g,f} & \downarrow G(g \circ f) \\ F(Z) & \xrightarrow{\mu_Z} & G(Z) \end{array} = \begin{array}{ccc} F(X) & \xrightarrow{\mu_X} & G(X) \\ \downarrow F(g \circ f) & \searrow \mu(g \circ f) & \downarrow G(g \circ f) \\ F(Z) & \xrightarrow{\mu_Z} & G(Z) \end{array}$$

- (ii) (**Unitality**) The following pasting diagrams are equal:

$$\begin{array}{ccc} F(X) & \xrightarrow{\mu_X} & G(X) \\ \downarrow F(\text{id}_X) & \searrow \psi_X & \downarrow \text{id}_{F(X)} \\ F(X) & \xrightarrow{\mu_X} & G(X) \end{array} = \begin{array}{ccc} F(X) & \xrightarrow{\mu_X} & G(X) \\ \downarrow F(\text{id}_X) & \searrow \mu(\text{id}_X) & \downarrow \text{id}_{F(X)} \\ F(X) & \xrightarrow{\mu_X} & G(X) \end{array}$$

Definition 1.6 (Modification). Continue with the same notation as above. Let $\mu, \nu : (F, \phi, \psi) \rightarrow (G, \varphi, \vartheta)$ be two pseudotransforms, a *modification* $\lambda : \mu \Rightarrow \nu$ is a collection of 2-cells $\lambda_X : \mu_X \Rightarrow \nu_X$ for each 0-cell $X \in \mathcal{A}$, such that the following two pasting diagrams are equal, for each 1-cell $f : X \rightarrow Y$ in $\mathcal{A}$:

$$\begin{array}{ccc} F(X) & \xrightarrow{\mu_X} & G(X) \\ \downarrow Ff & \searrow \lambda_X & \downarrow Gf \\ F(Y) & \xrightarrow{\mu_Y} & G(Y) \end{array} = \begin{array}{ccc} F(X) & \xrightarrow{\mu_X} & G(X) \\ \downarrow Ff & \searrow \mu_f & \downarrow Gf \\ F(Y) & \xrightarrow{\mu_Y} & G(Y) \end{array}$$

1.3. G -Categories.

Definition 1.7. A small category $\mathcal{A}$ is a G -category, if the object and morphism sets $\text{Ob}(\mathcal{A}), \text{Mor}(\mathcal{A})$ are both (left) G -sets, such that the following four structure maps (S, T, I, C) (stands for *source, target, identity, and composition*) are all G -equivariant:

- $S, T : \text{Mor}(\mathcal{A}) \rightarrow \text{Ob}(\mathcal{A})$ by $S(X \xrightarrow{f} Y) := X$, and $T(X \xrightarrow{f} Y) := Y$.
- $I : \text{Ob}(\mathcal{A}) \rightarrow \text{Mor}(\mathcal{A})$ by $I(X) := \text{id}_X$.
- $C : \text{Mor}(\mathcal{A}) \times_{S,T} \text{Mor}(\mathcal{A}) \rightarrow \text{Mor}(\mathcal{A})$ by $C(g, f) := g \circ f$.

Here, $\text{Mor}(\mathcal{A}) \times_{S,T} \text{Mor}(\mathcal{A}) := \{(g, f) \in \text{Mor}(\mathcal{A})^2 \mid S(g) = T(f)\}$, precisely the pair of morphisms that're composable, and it endows G -set structure via restricting the diagonal G -action on $\text{Mor}(\mathcal{A})^2$. This is well-defined due to the assumption S, T are G -maps.

Another equivalent characterization can be viewed as a group homomorphism $G \rightarrow \text{Aut}_{\mathbf{Cat}}(\mathcal{A})$, which assigns each $g \in G$ an automorphism functor on $\mathcal{A}$ such that composition respects G 's multiplication structure, and identity gets sent to the identity functor. So, from now on we view each element $g \in G$ as a functor, whenever a category is specified as a G -category.

Definition 1.8. Let $\mathbf{Cat}_G$ denotes the 2-category of small G -categories, with (nonequivariant) functors as 1-cells, natural transformations as 2-cells. The hom-category between any two G -categories $\mathbf{Cat}_G(\mathcal{A}, \mathcal{B})$ also has a natural G -action via conjugation. Which, the G -fixed subcategory of the hom-category $[\mathbf{Cat}_G(\mathcal{A}, \mathcal{B})]^G$ precisely describes all G -functor, or the functors $F : \mathcal{A} \rightarrow \mathcal{B}$ such that $g \circ F = F \circ g$ for all $g \in G$.

Another important notion is the twisted G -action on products when given a group homomorphism $\alpha : G \rightarrow \Sigma_n$. Given a category $\mathcal{A}$, the n -fold product $\mathcal{A}^n$ has a natural left Σ_n -action via permuting the coordinates; if $\mathcal{A}$ is a G -category, $\mathcal{A}^n$ also endows a natural diagonal G -action. The two coordinates to give a natural left $(G \times \Sigma_n)$ -action on $\mathcal{A}^n$.

Definition 1.9. Let $\mathcal{A}^{n^\alpha}$ be a G -category with the underlying category being $\mathcal{A}^n$, with the G -action defined as follow:

$$g \cdot_\alpha (X_1, \dots, X_n) := (g \cdot X_{\alpha(g)^{-1}(1)}, \dots, g \cdot X_{\alpha(g)^{-1}(n)})$$

Another characterization can be understood as follow: Given α 's graph subgroup $\Gamma_\alpha := \{(g, \alpha(g)) \in G \times \Sigma_n\}_{g \in G}$, the projection $\pi_G : G \times \Sigma_n \rightarrow G$ restricts to an isomorphism $\Gamma_\alpha \xrightarrow{\sim} G$. The α -twisted action is induced by the pullback along this isomorphism.

1.4. Chaotic-Forgetful Adjunction. Given any small category $\mathcal{A} \in \mathbf{Cat}$, its object $\text{Ob}(\mathcal{A})$ forms a set; moreover, any functor $F : \mathcal{A} \rightarrow \mathcal{B}$ also has an underlying map of objects $F : \text{Ob}(\mathcal{A}) \rightarrow \text{Ob}(\mathcal{B})$. Conversely, given a set, it has certain category structure:

Definition 1.10. Given a set X , define $\mathcal{E}X$ to be the *chaotic category* with underlying object set $\text{Ob}(\mathcal{E}X) := X$, by encoding a unique isomorphism $x \xrightarrow{\sim} y$ between any element $x, y \in X$. The composition

$$(1.11) \quad (x \xrightarrow{\sim} y \xrightarrow{\sim} z) = (x \xrightarrow{\sim} z)$$

using the uniqueness.

Given a set map $f : X \rightarrow Y$, define the functor $\mathcal{E}f : \mathcal{E}X \rightarrow \mathcal{E}Y$ so that the underlying object map is f , and for each morphism $x \xrightarrow{\sim} y$ in $\mathcal{E}X$, $\mathcal{E}f(x \rightarrow y) := f(x) \rightarrow f(y)$, the unique morphism between $f(x), f(y)$ in $\mathcal{E}Y$.

Inspecting the definition, we see $\mathcal{E} : \mathbf{Set} \rightarrow \mathbf{Cat}$ is a well-defined functor, called the *chaotic functor*.

Notice if a functor $F : \mathcal{A} \rightarrow \mathcal{B}$ has a chaotic target $\mathcal{B}$, then all the morphisms $f : X \rightarrow Y$ in $\mathcal{A}$ must be sent to the unique morphism $F(X) \rightarrow F(Y)$ in $\mathcal{B}$, hence F is dictated by the object map $F : \text{Ob}(\mathcal{A}) \rightarrow \text{Ob}(\mathcal{B})$. Such uniqueness specifies an adjunction $(\text{Ob} \dashv \mathcal{E})$.

The consequence is the following natural isomorphism of homset for any two sets X, Y :

$$(1.12) \quad \mathbf{Set}(X, Y) = \mathbf{Set}(\text{Ob}(\mathcal{E}X), Y) \cong \mathbf{Cat}(\mathcal{E}X, \mathcal{E}Y)$$

Remark 1.13. Let $\mathbf{Set}_G$ denotes the category of (left) G -sets, with all maps between G -sets (not necessarily equivariant) being the morphisms, the above adjunction can be promoted to $\text{Ob} : \mathbf{Cat}_G \rightarrow \mathbf{Set}_G$, and $\mathcal{E} : \mathbf{Set}_G \rightarrow \mathbf{Cat}_G$. Also, similar notions hold when transfer from left to right G -sets/categories.

In this exposition we only need $\mathcal{E}$ to be a right adjoint, there are richer details demonstrated in [?],

2. OPERADS, OPERADIC ALGEBRA, AND OPERAD $\mathcal{P}$

2.1. Symmetric Monoidal Categories.

Definition 2.1. A category $\mathcal{V}$ is a *symmetric monoidal category*, if there is a bifunctor $\otimes : \mathcal{V} \times \mathcal{V} \rightarrow \mathcal{V}$ (called *monoidal product*), and an object $\kappa \in \mathcal{V}$ (called *unit/monoidal unit*), together with the following natural isomorphisms:

- The *associator* $a : (-_1 \otimes -_2) \otimes -_3 \xrightarrow{\sim} -_1 \otimes (-_2 \otimes -_3)$.
- The *left unit* $\lambda : \kappa \otimes - \xrightarrow{\sim} \text{Id}_{\mathcal{V}}$ and the *right unit* $\rho : - \otimes \kappa \xrightarrow{\sim} \text{Id}_{\mathcal{V}}$.
- The *Symmetric Braiding* $B : -_1 \otimes -_2 \xrightarrow{\sim} -_2 \otimes -_1$, with extra property $B_{Y,X} : Y \otimes X \rightarrow X \otimes Y$ equals to $B_{X,Y}^{-1} : X \otimes Y \rightarrow Y \otimes X$.

They also satisfy the following axioms:

- **(Triangle)**

$$\begin{array}{ccc} (X \otimes \kappa) \otimes Y & \xrightarrow{a_{X,\kappa,Y}} & X \otimes (\kappa \otimes Y) \\ & \searrow \rho_X \otimes \text{id}_Y \quad \swarrow \text{id}_X \otimes \lambda_Y & \\ & X \otimes Y & \end{array}$$

- **(Pentagon)**

$$\begin{array}{ccc} ((X \otimes Y) \otimes Z) \otimes W & \xrightarrow{a_{X \otimes Y, Z, W}} & (X \otimes Y) \otimes (Z \otimes W) \\ \downarrow a_{X,Y,Z} \otimes \text{id}_W & & \downarrow a_{X,Y,Z \otimes W} \\ (X \otimes (Y \otimes Z)) \otimes W & & X \otimes (Y \otimes (Z \otimes W)) \\ \searrow a_{X,Y \otimes Z, W} & \nearrow \text{id}_X \otimes a_{Y,Z,W} & \\ & X \otimes ((Y \otimes Z) \otimes W) & \end{array}$$

- **(Hexagon)**

$$\begin{array}{ccccc} (X \otimes Y) \otimes Z & \xrightarrow{a_{X,Y,Z}} & X \otimes (Y \otimes Z) & \xrightarrow{B_{X,Y \otimes Z}} & (Y \otimes Z) \otimes X \\ \downarrow B_{X,Y} \otimes \text{id}_Z & & & & \downarrow a_{Y,Z,X} \\ (Y \otimes X) \otimes Z & \xrightarrow{a_{Y,X,Z}} & Y \otimes (X \otimes Z) & \xrightarrow{\text{id}_Y \otimes B_{X,Z}} & Y \otimes (Z \otimes X) \end{array}$$

Based on *MacLane's Coherence Theorem*, the above two axioms guarantees all finite sequences of a, λ, ρ and their inverses agree, namely there is a unique isomorphism $P_1 \rightarrow P_2$, where P_i are (possibly) different orders of performing product on $X_1, \dots, X_n \in \mathcal{V}$. For details, cf. [?].

The category of sets **Set**, small categories **Cat** in 1.3, and small G -categories **Cat_G** in 1.8 are central examples for this, where the cartesian product $\mathcal{A} \times \mathcal{B}$ serves as monoidal product, and the terminal set/category $*$ serves as units for both.

2.2. Operads, Operadic Algebras.

Definition 2.2. Given symmetric monoidal category $\mathcal{V}$, a Σ *operad* $\mathcal{C}$ in $\mathcal{V}$ is a collection of objects $\{\mathcal{C}(j) \in \mathcal{V}\}_{j \in \mathbb{N}}$, each $\mathcal{C}(j)$ with a right Σ_j -action (a group

homomorphism $\Sigma_j^{\text{op}} \rightarrow \text{Aut}_{\mathcal{V}}(\mathcal{C}(j))$, a *unit map* $\eta : 1 \rightarrow \mathcal{C}(1)$, together with composition maps

$$\gamma : \mathcal{C}(k) \otimes \mathcal{C}(j_1) \otimes \dots \otimes \mathcal{C}(j_k) \rightarrow \mathcal{C}(j_1 + \dots + j_k)$$

that satisfies the following axioms:

- *Unit:*

$$\begin{array}{ccc} \mathcal{C}(j) \otimes \kappa^{\otimes j} & \xrightarrow{\sim} & \mathcal{C}(j) \\ \searrow \text{id}_{\mathcal{C}(j)} \otimes \eta^{\otimes j} & & \nearrow \gamma \\ & \mathcal{C}(j) \otimes \mathcal{C}(1)^{\otimes j} & \end{array} \quad , \quad \begin{array}{ccc} \kappa \otimes \mathcal{C}(j) & \xrightarrow{\sim} & \mathcal{C}(j) \\ \searrow \eta \otimes \text{id}_{\mathcal{C}(j)} & & \nearrow \gamma \\ & \mathcal{C}(1) \otimes \mathcal{C}(j) & \end{array}$$

- *Associativity:* Fix natural numbers $k, j = \sum_{s=1}^k j_s$, define index $g_s := j_1 + \dots + j_s$ for $1 \leq s \leq k$. For each $i_1, \dots, i_{g_k} = i_j$, with sum $i = \sum_{r=1}^j i_r$, and $h_s := i_{g_{(s-1)}+1} + \dots + i_{g_s}$ (so $i = \sum_{s=1}^k h_s$ also), the following diagram commutes:

$$\begin{array}{ccc} \left[\mathcal{C}(k) \otimes \left(\bigotimes_{s=1}^k \mathcal{C}(j_s) \right) \right] \otimes \left(\bigotimes_{r=1}^j \mathcal{C}(i_r) \right) & \xrightarrow{\gamma \otimes \text{id}} & \mathcal{C}(j) \otimes \left(\bigotimes_{r=1}^j \mathcal{C}(i_r) \right) \\ \downarrow \text{shuffle} & & \downarrow \gamma \\ \mathcal{C}(k) \otimes \left[\bigotimes_{s=1}^k \left(\mathcal{C}(j_s) \otimes \left(\bigotimes_{q=1}^{j_s} \mathcal{C}(i_{g_{(s-1)}+q}) \right) \right) \right] & \xrightarrow{\text{id}_{\mathcal{C}(k)} \otimes (\bigotimes_s \gamma)} & \mathcal{C}(k) \otimes \left(\bigotimes_{s=1}^k \mathcal{C}(h_s) \right) \\ & & \uparrow \gamma \\ & & \mathcal{C}(i) \end{array}$$

- *Equivariance:*

- Given $\sigma \in \Sigma_k$, $j_1 + \dots + j_k = \Sigma_j$, let $\sigma(j_1, \dots, j_k) \in \Sigma_j$ denotes “permutation of k -blocks of letters”, by $(j_1, \dots, j_k) \mapsto (j_{\sigma^{-1}(1)}, \dots, j_{\sigma^{-1}(k)})$. Then, the following diagram commutes:

$$\begin{array}{ccc} \mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^k \mathcal{C}(j_r) \right) & \xrightarrow{\sigma \otimes \sigma^{-1}} & \mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^k \mathcal{C}(j_{\sigma(r)}) \right) \\ \gamma \downarrow & & \downarrow \gamma \\ \mathcal{C}(j) & \xrightarrow{\sigma(j_{\sigma(1)}, \dots, j_{\sigma(k)})} & \mathcal{C}(j) \end{array}$$

- Given $\tau_r \in \Sigma_{j_r}$, denote $\tau_1 \oplus \dots \oplus \tau_k \in \Sigma_j$ by letting each τ_r permuting the j_r th block in j . Then, the following diagram commutes:

$$\begin{array}{ccc} \mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^k \mathcal{C}(j_r) \right) & \xrightarrow{\text{id}_{\mathcal{C}(k)} \otimes (\bigotimes_r \tau_r)} & \mathcal{C}(k) \otimes \left(\bigotimes_{r=1}^k \mathcal{C}(j_r) \right) \\ \gamma \downarrow & & \downarrow \gamma \\ \mathcal{C}(j) & \xrightarrow{\tau_1 \oplus \dots \oplus \tau_k} & \mathcal{C}(j) \end{array}$$

If $\mathcal{C}(0) = \kappa$ (with $\gamma : \mathcal{C}(0) \rightarrow \mathcal{C}$ being $\text{id}_{\mathcal{C}(0)}$), the operad is *reduced*.

Definition 2.3. Given an operad $\mathcal{C}$ over a symmetric monoidal category $\mathcal{V}$, a $\mathcal{C}$ -algebra is an object $A \in \mathcal{V}$, with collections of morphisms $\theta : \mathcal{C}(j) \otimes A^j \rightarrow A$, that satisfies the following axioms:

- *Associativity:* Given $j = \sum_{i=1}^k j_i$, the following diagram commutes:

$$\begin{array}{ccc}
 \left[\mathcal{C}(k) \otimes \left(\bigotimes_{i=1}^k \mathcal{C}(j_i) \right) \right] \otimes A^{\otimes j} & \xrightarrow{\gamma \otimes \text{id}_{A^{\otimes j}}} & \mathcal{C}(j) \otimes A^{\otimes j} \\
 \downarrow \text{shuffle} & & \downarrow \theta \\
 \mathcal{C}(k) \otimes \left[\bigotimes_{i=1}^k (\mathcal{C}(j_i) \otimes A^{\otimes j_i}) \right] & \xrightarrow{\text{id}_{\mathcal{C}(k)} \otimes \theta^k} & \mathcal{C}(k) \otimes A^{\otimes k} \\
 & & \uparrow \theta \\
 & & A
 \end{array}$$

(Note: This part uses the isomorphism $A^{\otimes j} \cong \bigotimes_{i=1}^k A^{\otimes j_i}$).

- *Unit:* The following diagram commutes:

$$\begin{array}{ccc}
 \kappa \otimes A & \xrightarrow{\sim} & A \\
 \searrow \eta \otimes \text{id}_A & & \nearrow \theta \\
 & \mathcal{C}(1) \otimes A &
 \end{array}$$

- *Eauivariance:* Given any permutation $\sigma \in \Sigma_j$, the following diagram commutes:

$$\begin{array}{ccc}
 \mathcal{C}(j) \otimes A^{\otimes j} & \xrightarrow{\sigma \otimes \sigma^{-1}} & \mathcal{C}(j) \otimes A^{\otimes j} \\
 \searrow \theta & & \swarrow \theta \\
 & A &
 \end{array}$$

Example 2.4. The *associativity operad* $\mathcal{M}$ in **Set** is defined as $\mathcal{M}(j) := \Sigma_j$ (the symmetric group for cardinality j), which endows a natural right Σ_j -action via right multiplication. The operadic structure map is given by

(2.5)

$$\gamma : \Sigma_k \times \prod_{r=1}^k \Sigma_{j_r} \rightarrow \Sigma_j, \quad \gamma(\sigma; \tau_1, \dots, \tau_k) := \tau_{\sigma^{-1}(1)} \oplus \dots \oplus \tau_{\sigma^{-1}(k)} \circ \sigma(j_{\sigma(1)}, \dots, j_{\sigma(k)})$$

where $j = \sum_r j_r$, $\sigma(j_1, \dots, j_k) \in \Sigma_j$ denotes the block permutation of $j_1, \dots, j_k$ within j (in order), and $\tau_1 \oplus \dots \tau_k$ is the image of the embedding $\oplus : \prod_{r=1}^k \Sigma_{j_r} \rightarrow \Sigma_j$, by viewing the r th block as finite set of cardinality j_r . Inspecting the definitions, the algebra over $\mathcal{M}$ precisely describes the set theoretic monoids. For more details, cf. [?] Section 4.1.

2.3. Permutativity Operad $\mathcal{P}$. Now, recall that the chaotic functor $\mathcal{E} : \mathbf{Set} \rightarrow \mathbf{Cat}$ is a right adjoint of $\text{Ob} : \mathbf{Cat} \rightarrow \mathbf{Set}$, in particular it preserves products. Hence, the chaotic categorification $\mathcal{P}(j) := \mathcal{E}\mathcal{M}(j)$ also has a natural operadic structure, via the identification

$$(2.6) \quad \mathcal{E}\mathcal{M}(k) \times \prod_{r=1}^k \mathcal{E}\mathcal{M}(j_r) \cong \mathcal{E} \left(\mathcal{M}(k) \times \prod_{r=1}^k \mathcal{M}(j_r) \right) \xrightarrow{\mathcal{E}\gamma} \mathcal{E}\mathcal{M}(j).$$

This is called the *permutativity operad in $\mathbf{Cat}$* , or the *Barratt-Eccles operad*. As an abuse of notation, the operadic structure map of $\mathcal{P}$ is also denoted γ .

3. G -OPERAD $\mathcal{P}_G$ AND ITS PSEUDOALGEBRA

3.1. The G -Operad $\mathcal{P}_G$. If view each $\mathcal{P}(j)$ as a G -trivial G -category, it's also an operad in $\mathbf{Cat}_G$. Its underlying algebra is called the *naive permutative G -category*, as specified in [?] Section 4.1, it produces a naive G -spectrum under equivariant infinite loop space theory. In the same source Section 4.2, a richer equivariant structure is introduced which produces the *equivariant Barratt Eccles G -operad* $\mathcal{P}_G$ as follow:

Definition 3.1. The j th category $\mathcal{P}_G(j) := \mathbf{Cat}(\mathcal{E}G, \mathcal{P}(j))$, with the operadic structure map determined by the fact that $\mathbf{Cat}(\mathcal{E}G, -)$ preserves product:

$$(3.2) \quad \mathbf{Cat}(\mathcal{E}G, \mathcal{P}(k)) \times \prod_{r=1}^k \mathbf{Cat}(\mathcal{E}G, \mathcal{P}(j_r)) \cong \mathbf{Cat} \left(\mathcal{E}G, \mathcal{P}(k) \times \prod_{r=1}^k \mathcal{P}(j_r) \right) \xrightarrow{\gamma \circ -} \mathbf{Cat}(\mathcal{E}, \mathcal{P}(j))$$

Abuse of notation, we'll denote the operadic structure map as γ again.

Another formulation can be done by defining $P_G(j) := \mathbf{Set}(G, \Sigma_j)$ in $\mathbf{Set}$, then utilize chaotic functor promoting this to a chaotic operad $\mathcal{E}\mathbf{Set}(G, \Sigma) \cong \mathcal{P}_G$.

3.2. The 2-Category $\mathcal{P}_G$ -PsAlg. The general notion of operadic pseudoalgebra is defined in [?], [?], and [?]. Here we only take the case when the operad is $\mathcal{P}_G$, together with some description of G -equivariance.

Definition 3.3 ($\mathcal{P}_G$ -pseudoalgebra). A $\mathcal{P}_G$ -pseudoalgebra is a G -category $\mathcal{A}$ and operad action functors

$$\theta_j : \mathcal{P}_G(j) \times \mathcal{A}^j \rightarrow \mathcal{A}$$

for each $n \in \mathbb{N}$, together with an invertible natural transformation ϕ_1 in the diagram

$$(3.4) \quad \begin{array}{ccc} \mathcal{A} & \xrightarrow{\iota} & \mathcal{P}_G(1) \times \mathcal{A} \\ & \searrow \phi_1 & \downarrow \theta_1 \\ & \text{Id}_{\mathcal{A}} & \mathcal{C} \end{array}$$

An invertible natural transformation $\phi = \phi(k; j_1, \dots, j_k)$ for each diagram

$$(3.5) \quad \begin{array}{ccc} \mathcal{P}_G(k) \times \left(\prod_{r=1}^k \mathcal{P}_G(j_r) \times \mathcal{A}^{j_r} \right) & \xrightarrow{\text{id} \times \prod_r \theta_{j_r}} & \mathcal{P}_G(k) \times \mathcal{A}^k \\ \downarrow \text{shuffle} & \searrow \phi & \downarrow \theta_k \\ \mathcal{P}_G(k) \times \prod_{r=1}^k \mathcal{P}_G(j_r) \times \mathcal{A}^{j_r} & \xrightarrow{\gamma \times \text{id}} & \mathcal{P}_G(j_+) \times \mathcal{A}^{j_+} \end{array}$$

such that the following strict equivariance hold for θ_j and ϕ :

- (i) **(Equivariance of θ_j)** The following two diagrams commutes for $\sigma \in \Sigma_j$ and $g \in G$:

(3.6)

$$\begin{array}{ccc}
 \mathcal{P}_G(j) \times \mathcal{A}^j & \xrightarrow{\sigma \times \sigma^{-1}} & \mathcal{P}_G(j) \times \mathcal{A}^j \\
 \searrow \theta_j & & \swarrow \theta_j \\
 & \mathcal{A} &
 \end{array}
 \qquad
 \begin{array}{ccc}
 \mathcal{P}_G(j) \times \mathcal{A}^j & \xrightarrow{g \times g^{-1}} & \mathcal{P}_G(j) \times \mathcal{A}^j \\
 \theta_j \downarrow & & \downarrow \theta_j \\
 \mathcal{A} & \xrightarrow{g^{-1}} & \mathcal{A}
 \end{array}$$

The left is the operadic equivariance, the right is G -equivariance by asserting a product G -action on $\mathcal{P}(j) \times \mathcal{A}^j$.

- (ii) **(Equivariance of ϕ)** The following equalities hold when given $\sigma \in \Sigma_k$, and $\tau_r \in \Sigma_{j_r}$:

$$\phi(k; j_1, \dots, j_k) = \phi(k; j_{\sigma(1)}, \dots, j_{\sigma(k)}) \circ (\sigma \times \sigma^{-1})$$

$$\phi(k; j_1, \dots, j_k) = \phi(k; j_1, \dots, j_k) \circ (\text{id} \times \prod_r (\tau_r \times \tau_r^{-1}))$$

They also need to satisfy the following coherence properties:

- (iii) **(Operadic Composition)** The following pasting diagrams are equal, given $j_+ = \sum j_r$, $i_r = \sum_s i_{r,s}$, and $i_+ = \sum_{r,s} i_{r,s}$ with $1 \leq r \leq k$ and $1 \leq s \leq j_r$:

$$\begin{array}{ccccc}
 \mathcal{P}_G(k) \times \prod_r (\mathcal{P}_G(j_r) \times \prod_s (\mathcal{P}_G(i_{r,s}) \times \mathcal{A}^{i_{r,s}})) & \xrightarrow{\text{id} \times \prod (\text{id} \times \theta_{i_{r,s}})} & \mathcal{Q}_G(k) \times \prod_r (\mathcal{P}_G(j_r) \times \mathcal{A}^{j_r}) & \xrightarrow{\text{id} \times \prod \theta_{j_r}} & \mathcal{P}_G(k) \times \mathcal{A}^k \\
 \downarrow \gamma \times \text{id} & & \downarrow \gamma & \searrow \phi & \downarrow \theta_k \\
 \mathcal{P}_G(j_+) \times \prod_{r,s} (\mathcal{P}_G(i_{r,s}) \times \mathcal{A}^{i_{r,s}}) & \xrightarrow{\text{id} \times \prod \theta_{i_{r,s}}} & \mathcal{P}_G(j_+) \times \mathcal{A}^{j_+} & \xrightarrow{\theta_{j_+}} & \mathcal{A} \\
 \searrow \gamma \times \text{id} & & \downarrow \phi & \nearrow \theta_{j_+} & \\
 & & \mathcal{P}_G(i_+) \times \mathcal{A}^{i_+} & \xrightarrow{\theta_{i_+}} & \mathcal{A}
 \end{array}$$

equals

$$\begin{array}{ccccc}
 \mathcal{P}_G(k) \times \prod_r (\mathcal{P}_G(j_r) \times \prod_s (\mathcal{P}_G(i_{r,s}) \times \mathcal{A}^{i_{r,s}}))^{\text{id} \times \prod (\text{id} \times \prod \theta_{i_{r,s}})} & \xrightarrow{\quad} & \mathcal{P}_G(k) \times \prod_r (\mathcal{P}_G(j_r) \times \mathcal{A}^{j_r}) & & \\
 \downarrow \gamma \times \text{id} & \searrow \text{id} \times \prod (\gamma \times \text{id}) & \downarrow \text{id} \times \prod \phi & \searrow \text{id} \times \prod \theta_{j_r} & \\
 \mathcal{P}_G(j_+) \times \prod_{r,s} (\mathcal{P}_G(i_{r,s}) \times \mathcal{A}^{i_{r,s}}) & & \mathcal{P}_G(k) \times \prod_r (\mathcal{P}_G(i_r) \times \mathcal{A}^{i_r}) & \xrightarrow{\text{id} \times \prod \gamma} & \mathcal{P}_G(k) \times \mathcal{A}^k \\
 & \searrow \gamma \times \text{id} & \downarrow \gamma \times \text{id} & \nearrow \phi & \downarrow \theta_k \\
 & & \mathcal{P}_G(i_+) \times \mathcal{A}^{i_+} & \xrightarrow{\theta_{i_+}} & \mathcal{A}
 \end{array}$$

The commuting square above is due to associativity of operad.

- (iv) **(Operadic Identity)** The following two pairs of two pasting diagrams are equal:

$$\begin{array}{ccc}
 \mathcal{P}_G(k) \times \mathcal{A}^k & \xrightarrow{\text{id} \times \iota^k} & \mathcal{P}_G(k) \times (\mathcal{P}_G(1) \times \mathcal{A})^k \xrightarrow{\text{id} \times \theta_1^k} \mathcal{P}_G(k) \times \mathcal{A}^k \\
 & \downarrow \text{shuffle} & \downarrow \theta_k \\
 & \mathcal{P}_G(k) \times \mathcal{P}_G(1)^k \times \mathcal{A}^k & \nearrow \phi \\
 & \downarrow \gamma \times \text{id} & \downarrow \theta_k \\
 & \mathcal{P}_G(k) \times \mathcal{A}^k & \xrightarrow{\theta_k} \mathcal{A}
 \end{array}$$

equals

$$\begin{array}{ccccc}
 & & \mathcal{P}_G(k) \times (\mathcal{P}_G(1) \times \mathcal{A})^k & & \\
 & \nearrow \text{id} \times \iota^k & \downarrow \text{id} \times \prod \phi & \searrow \text{id} \times \theta_1^k & \\
 \mathcal{P}_G(k) \times \mathcal{A}^k & \xrightarrow{\quad} & \mathcal{P}_G(k) \times \mathcal{A}^k & \xrightarrow{\theta_k} & \mathcal{A} \\
 \downarrow \text{id} \times \iota^k & \searrow \iota^k & \nearrow \gamma \times \text{id} & & \\
 \mathcal{P}_G(k) \times (\mathcal{P}_G(1) \times \mathcal{A})^k & \xrightarrow{\text{shuffle}} & \mathcal{P}_G(k) \times \mathcal{P}_G(1)^k \times \mathcal{A}^k & &
 \end{array}$$

(The two triangles right above commutes due to the unit axioms of operad).

Also:

$$\begin{array}{ccccc}
 \mathcal{P}_G(k) \times \mathcal{A}^k & \xrightarrow{\iota} & \mathcal{P}_G(1) \times \mathcal{P}_G(k) \times \mathcal{A}^k & \xrightarrow{\text{id} \times \theta_k} & \mathcal{P}_G(1) \times \mathcal{A} \\
 & \searrow \text{id} & \downarrow \gamma \times \text{id} & \nearrow \phi & \downarrow \theta_1 \\
 & & \mathcal{P}_G(k) \times \mathcal{A}^k & \xrightarrow{\theta_k} & \mathcal{A}
 \end{array}$$

equals

$$\begin{array}{ccc}
 \mathcal{P}_G(k) \times \mathcal{A}^k & \xrightarrow{\iota} & \mathcal{P}_G(1) \times \mathcal{P}_G(k) \times \mathcal{A}^k \\
 \downarrow \theta_k & & \downarrow \text{id} \times \theta_k \\
 \mathcal{A} & \xrightarrow{\iota} & \mathcal{P}_G(1) \times \mathcal{A} \\
 & \searrow \phi_1 & \downarrow \theta_1 \\
 & \text{id} & \mathcal{A}
 \end{array}$$

The first diagram's triangle commutes due to the operadic unit axiom.

Remark 3.7. In [?] Corner and Gurski includes unital natural isomorphisms, while in [?] it requires strict unitality. The main result in the nonequivariant case of [?] alleviates the result from strict unitality.

Definition 3.8 ($\mathcal{P}_G$ -pseudomorphism). A $\mathcal{P}_G$ -pseudomorphism $(\mathbb{F}, \zeta) : (\mathcal{A}, \theta, \phi) \rightarrow (\mathcal{B}, \vartheta, \varphi)$ consists of a functor $\mathbb{F} : \mathcal{A} \rightarrow \mathcal{B}$, such that for each j in the following diagram commutes up to an invertible natural transformation ζ_j :

$$\begin{array}{ccc}
 \mathcal{P}_G(j) \times \mathcal{A}^j & \xrightarrow{\text{id} \times \mathbb{F}^j} & \mathcal{P}_G(j) \times \mathcal{B}^j \\
 \downarrow \theta_j & \searrow \zeta_j & \downarrow \vartheta_j \\
 \mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B}
 \end{array}$$

Moreover, they satisfy the following properties:

- (i) **(Equivariance of ζ)** For each $j, \zeta_j = \zeta_j \circ (\sigma \times \sigma^{-1})$ for all $\sigma \in \Sigma_j$. **Check G -equivariance**
- (ii) **(Preserves Identity)** The following two pasting diagrams are equal:

$$\begin{array}{ccc}
 \mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B} \\
 \downarrow \iota & & \downarrow \iota \\
 \mathcal{P}_G(1) \times \mathcal{A} & \xrightarrow{\text{id} \times \mathbb{F}} & \mathcal{P}_G(1) \times \mathcal{B} \\
 \downarrow \theta_1 & \searrow \zeta_1^{-1} & \downarrow \vartheta_1 \\
 \mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B}
 \end{array}
 \quad \text{with a curved arrow } \text{id} \text{ from } \mathcal{B} \text{ to } \mathcal{B}$$

$$=
 \begin{array}{ccc}
 \mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B} \\
 \downarrow \iota & \searrow \text{id} & \downarrow \text{id} \\
 \mathcal{P}_G(1) \times \mathcal{A} & \xrightarrow{\theta_1} & \mathcal{A} \\
 \downarrow \theta_1 & \searrow \phi_1 & \downarrow \theta_1 \\
 \mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B}
 \end{array}$$

(iii) **(Preserves Composition)** The following two pasting diagrams are equal:

$$\begin{array}{ccccc}
 \mathcal{P}_G(k) \times \prod_r (\mathcal{P}_G(j_r) \times \mathcal{A}^{j_r}) & \xrightarrow{\text{id} \times \prod (\text{id} \times \mathbb{F}^{j_r})} & \mathcal{P}_G(k) \times \prod_r (\mathcal{P}_G(j_r) \times \mathcal{B}^{j_r}) & & \\
 \downarrow \gamma & \searrow \text{id} \times \prod \theta_{j_r} & \swarrow \text{id} \times \prod \zeta_{j_r} & \searrow \text{id} \times \prod \vartheta_{j_r} & \\
 & \mathcal{P}_G(k) \times \mathcal{A}^k & \xrightarrow{\text{id} \times \mathbb{F}^k} & \mathcal{P}_G(k) \times \mathcal{B}^k & \\
 & \downarrow \theta_k & \swarrow \zeta_k & \downarrow \vartheta_k & \\
 \mathcal{P}_G(j) \times \mathcal{A}^j & \xleftarrow{\phi} & \mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B} \\
 & \searrow \theta_j & & &
 \end{array}$$

equals

$$\begin{array}{ccccc}
 \mathcal{P}_G(k) \times \prod_r (\mathcal{P}_G(j_r) \times \mathcal{A}^{j_r}) & \xrightarrow{\text{id} \times \prod (\text{id} \times \mathbb{F}^{j_r})} & \mathcal{P}_G(k) \times \prod_r (\mathcal{P}_G(j_r) \times \mathcal{B}^{j_r}) & & \\
 \downarrow \gamma & & \downarrow \gamma & \searrow \text{id} \times \prod \theta_{j_r} & \\
 & \mathcal{P}_G(j) \times \mathcal{A}^j & \xrightarrow{\text{id} \times \mathbb{F}^j} & \mathcal{P}_G(j) \times \mathcal{B}^j & \\
 & \searrow \theta_j & \swarrow \zeta_j & \searrow \vartheta_j & \\
 & \mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B} &
 \end{array}$$

Definition 3.9 ($\mathcal{P}_G$ -2-cells). Let $(\mathbb{F}, \zeta), (\mathbb{G}, \xi) : (\mathcal{A}, \theta, \phi) \rightarrow (\mathcal{B}, \vartheta, \varphi)$ be two $\mathcal{P}_G$ -pseudomorphisms, a $\mathcal{P}_G$ -2-cell $\lambda : (\mathbb{F}, \zeta) \Rightarrow (\mathbb{G}, \xi)$ is a natural transformation $\lambda : \mathbb{F} \rightarrow \mathbb{G}$, such that the following two pasting diagrams are equal:

$$\begin{array}{ccc}
 \mathcal{P}_G(j) \times \mathcal{A}^j & \xrightarrow{\text{id} \times \mathbb{F}^j} & \mathcal{P}_G(j) \times \mathcal{B}^j \\
 \downarrow \theta_j & \searrow \xi_j & \downarrow \vartheta_j \\
 \mathcal{A} & \xrightarrow{\mathbb{G}} & \mathcal{B}
 \end{array}
 =
 \begin{array}{ccc}
 \mathcal{P}_G(j) \times \mathcal{A}^j & \xrightarrow{\text{id} \times \mathbb{F}^j} & \mathcal{P}_G(j) \times \mathcal{B}^j \\
 \downarrow \theta_j & \searrow \zeta_j & \downarrow \vartheta_j \\
 \mathcal{A} & \xrightarrow[\lambda]{\mathbb{F}} & \mathcal{B}
 \end{array}$$

4. CATEGORY $\mathcal{F}_G$ AND ITS PSEUDOALGEBRAS

Within the equivariant infinite loop space machine, the purpose for $\mathcal{F}_G$ -pseudoalgebras (after strictifying to algebras) are to construct $\mathcal{F}_G$ -spaces, which are collections of G -spaces indexed over finite based G -sets (and morphisms between them encoding extra structure). The $\mathcal{F}_G$ -spaces are primary inputs in producing a genuine G -spectra, and the other main character we wish to study the correspondance. For details of $\mathcal{F}_G$ -spaces, cf. [?] Section 1 (which is named after Γ_G instead of $\mathcal{F}_G$).

The following definitions are equivariant generalizations of $\mathcal{F}$ -pseudofunctors in [?], and we'll utilize similar notations.

Definition 4.1. $\mathcal{F}_G$ denotes the *category of based finite sets*, where the objects are $\mathbf{n}^\alpha$, set wise being the finite set $\{0, 1, \dots, n\}$ with 0 fixed as the basepoint, and $\alpha : G \rightarrow \Sigma_n$ is a based G -action on the set.

The homset $\mathcal{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta)$ consists of all the based set map, which the homset also endows with a G -action via conjugation. From now on, we'll view $\mathcal{F}_G$ as a 2-category, where each hom-category $\mathcal{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta)$ is discrete with only identity 2-cells.

Furthermore, we define the following:

- $\Pi_G \subset \mathcal{F}_G$ denotes the subcategory where each morphism $\phi : \mathbf{n}^\alpha \rightarrow \mathbf{m}^\beta$ satisfies $|\phi^{-1}(j)| = 0, 1$ for all $j \neq 0$.
- $\Lambda_G, \Lambda_G^\perp, \Sigma_G \subset \Pi_G$ denotes the subcategory where each morphism is injective, surjective, and bijective respectively (in particular, $\Lambda_G \cap \Lambda_G^\perp = \Sigma_G$).
- $\mathcal{E}_G \subset \mathcal{F}_G$ denotes the subcategory where each morphism satisfies $\phi^{-1}(0) = \{0\}$.

We also denote the map $\phi_{\mathbf{n}^\alpha} : \mathbf{n}^\alpha \rightarrow \mathbf{1}$ (with $j \mapsto 1$ for all $j \neq 0$) in $\mathcal{E}$ as the *canonical product*.

4.1. 2-Category $\mathcal{F}_G$ -PsAlg.

Definition 4.2 ($\mathcal{F}_G$ -pseudoalgebra). An $\mathcal{F}_G$ -pseudoalgebra, is a pseudofunctor $\mathcal{B} : \mathcal{F}_G \rightarrow \mathbf{Cat}_G$ (cf. Definition 1.4), such that restricting to the subcategory $\Pi_G \subset \mathcal{F}_G$, $\mathcal{B}$ is a strict functor. Moreover, we require it to be reduced, namely $\mathcal{B}(0) = *$ (the terminal G -category), and each hom-functor

$$\mathcal{B}_{\mathbf{n}^\alpha, \mathbf{m}^\beta} : \mathcal{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta) \rightarrow \mathbf{Cat}_G(\mathcal{B}(\mathbf{n}^\alpha), \mathcal{B}(\mathbf{m}^\beta))$$

is a G -functor.

Let $\delta_i : \mathbf{n}^\alpha \rightarrow \mathbf{1}$ be the i th projection (namely $i \mapsto 1$, and any $i \neq j \mapsto 0$), since it belongs to Π_G , it generates a functor $\mathcal{B}\delta_i : \mathcal{B}(\mathbf{n}^\alpha) \rightarrow \mathcal{B}(\mathbf{1})$. Hence, by universality of product, the collection $\mathcal{B}\delta_i$ ($1 \leq i \leq n$) induces a functor $\delta : \mathcal{B}(\mathbf{n}^\alpha) \rightarrow \mathcal{B}(\mathbf{1})^{\mathbf{n}^\alpha}$ (where $\mathcal{B}(\mathbf{1})^{\mathbf{n}^\alpha} \cong \mathcal{B}(\mathbf{1})^{\mathbf{n}}$ nonequivariantly).

Definition 4.3 ($\mathcal{F}_{G, \text{vs}}$ and $\mathcal{F}_{G, \mathbb{R}}$ -pseudoalgebra). An $\mathcal{F}_G$ -pseudoalgebra $\mathcal{B}$ is *very special* (or an $\mathcal{F}_{G, \text{vs}}$ -pseudoalgebra), if $\delta : \mathcal{B}(\mathbf{n}^\alpha) \rightarrow \mathcal{B}(\mathbf{1})^{\mathbf{n}^\alpha}$ is an isomorphism for all $\mathbf{n}^\alpha \in \mathcal{F}_G$, and $\mathcal{B}$ is *strictly special* (or an $\mathcal{F}_{G, \mathbb{R}}$ -pseudoalgebra), if all δ is in fact identity.

Definition 4.4 ($\mathcal{F}_G$ -1-cell). Given $\mathcal{A}, \mathcal{B}$ two $\mathcal{F}_G$ -pseudoalgebras, a morphism (or $\mathcal{F}_G$ -1-cell) between them is a pseudotransform $(\mathbb{F}, \zeta) : \mathcal{A} \rightarrow \mathcal{B}$ (cf. Definition 1.5), such that restricting to Π_G , it becomes an actual natural transformation. Moreover, it respects G -equivariance for every $g \in G$ and morphism $f \in \text{Mor}(\mathcal{F}_G)$:

$$\zeta(g \circ f \circ g^{-1}) = g \circ \zeta(f) \circ g^{-1}$$

Definition 4.5 ($\mathcal{F}_G$ -2-cell). Given $\tilde{A}, \tilde{B}$ two $\mathcal{F}_G$ -pseudoalgebras, and two $\mathcal{F}_G$ -1-cells $(\mathbb{F}, \zeta), (\mathbb{G}, \xi) : \tilde{A} \rightarrow \tilde{B}$, an $\mathcal{F}_G$ -2-cell is a modification $\lambda : \mathbb{F} \Rightarrow \mathbb{G}$ (cf. Definition 1.6), such that for any morphism $f : \mathbf{n}^\alpha \rightarrow \mathbf{m}^\beta$ in $\mathcal{F}_G$, the following two pasting diagrams are equal:

$$\begin{array}{ccc}
 \begin{array}{ccc}
 & \mathbb{F}_{\mathbf{n}^\alpha} & \\
 \mathcal{A}(\mathbf{n}^\alpha) & \xRightarrow{\lambda} & \mathcal{B}(\mathbf{n}^\alpha) \\
 & \mathbb{G}_{\mathbf{n}^\alpha} & \\
 \downarrow \mathcal{A}f & \searrow \xi f & \downarrow \mathcal{B}f \\
 \mathcal{A}(\mathbf{m}^\beta) & \xrightarrow{\mathbb{G}_{\mathbf{m}^\beta}} & \mathcal{B}(\mathbf{m}^\beta)
 \end{array}
 & = &
 \begin{array}{ccc}
 \mathcal{A}(\mathbf{n}^\alpha) & \xrightarrow{\mathbb{F}_{\mathbf{n}^\alpha}} & \mathcal{B}(\mathbf{n}^\alpha) \\
 \downarrow \mathcal{B}f & \nearrow \zeta f & \downarrow \mathcal{B}f \\
 \mathcal{A}(\mathbf{m}^\beta) & \xRightarrow[\lambda]{\mathbb{F}_{\mathbf{m}^\beta}} & \mathcal{B}(\mathbf{m}^\beta) \\
 & \mathbb{G}_{\mathbf{m}^\beta} &
 \end{array}
 \end{array}$$

5. TRANSITION TO EQUIVARIANT CORRESPONDANCE

The eventual goal is constructing an $\mathcal{F}_{G,\mathbb{R}}$ -pseudoalgebra out of an $\mathcal{P}_G$ -pseudoalgebra, while extend it to a 2-functor. [?] provides a general construction for the non-equivariant case (or $G = \mathbf{1}$).

Given a $\mathcal{P}$ -pseudoalgebra $\mathcal{A}$, define object map $\mathcal{B} : \mathcal{F} \rightarrow \mathbf{Cat}$ by $\mathcal{B}(\mathbf{n}) := \mathcal{A}^n$. For extending to a pseudofunctor (and restricting to functor on Π), the central concepts include:

- (i) Substituting the vanishing coordinates of maps $\psi : \mathbf{n} \rightarrow \mathbf{m}$ in Π using the specified basepoint $\theta_0 : * \cong \mathcal{P}(0) \rightarrow \mathcal{A}$. In particular, since each $j \in \mathbf{m}$ has preimage being empty or a singleton, define $\psi : \mathcal{A}^n \rightarrow \mathcal{A}^m$ by

$$(5.1) \quad \psi(a_1, \dots, a_n) := (a_{\psi^{-1}(1)}, \dots, a_{\psi^{-1}(m)}),$$

with $a_{\psi^{-1}(j)} := *$ (the basepoint) if $\psi^{-1}(j) = \emptyset$. This defines $\mathcal{B}$ to be a functor with source Π .

- (ii) Treating each canonical n -fold product $\phi_n : \mathbf{n} \rightarrow \mathbf{1}$ in $\mathcal{E}$ (with $\phi_n(j) = 1$ for all $j \neq 0$) using the n -fold product $e \times \mathcal{A} \hookrightarrow \mathcal{P}(n) \times \mathcal{A} \xrightarrow{\gamma} \mathcal{A}$.
- (iii) Finally, show the decomposition laws, that maps in $\mathcal{E}$ are compositions of permutation and wedge of canonical products (when fixing a specific order, it's unique), while maps in $\mathcal{F}$ decomposes into a projection and a map in $\mathcal{E}$. Using the laws to construct functors $\mathcal{A}^n \rightarrow \mathcal{A}^m$ using each map ψ .

The coherent natural isomorphisms for $\mathcal{P}$ -pseudoalgebra provides the pseudo-functoriality modulo extensive diagram chasing.

Step (i) is generalizes naturally to the equivariant case, as each $\mathcal{P}_G$ -pseudoalgebra $\mathcal{A}$ has a natural basepoint specified by $\theta_0 : * \cong \mathcal{P}_G(0) \rightarrow \mathcal{A}$, each $\mathcal{A}^{\mathbf{n}^\alpha} \cong \mathcal{A}^n$ nonequivariantly, and the maps in Π_G includes all nonequivariant ones, hence the same construction carries over.

For (ii), each canonical product $\phi_{\mathbf{n}^\alpha} : \mathbf{n}^\alpha \rightarrow \mathbf{1}$ has a choice given by $\alpha \times \mathcal{A}^n \hookrightarrow \mathcal{P}_G(n) \times \mathcal{A}^n \xrightarrow{\theta_n} \mathcal{A}$ (by treating γ), with some equivariance details we'll specify below.

For (iii), the main problem for equivariant case is: Not every map can be expressed as wedges of canonical product, as the preimage of an element may not necessarily be wedges of G -orbits. So, potentially some choices

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6. BIBLIOGRAPHY

The bibliography should list all sources that you have used and referenced. And you should reference anything you use. Especially if you quote any result without proof, you MUST give a reference. And never ever should you copy material directly or more or less directly, from a source. If you include a figure or diagram from some source, that source should be identified.

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